Research_Question_1

2026-03-19

The data should already be stored in the environment so this document assumes you already have it

This chunk is responsible for printing the number of unique crime codes and their descriptions

```
library(readr)
df <- read_csv("Fixed_Date_version.csv")
```

```
## Rows: 965944 Columns: 28
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (12): Date Rptd, AREA NAME, Crm Cd Desc, Mocodes, Vict Sex, Vict Descen...
## dbl  (13): DR_NO, TIME OCC, AREA, Rpt Dist No, Part 1-2, Crm Cd, Vict Age, P...
## lgl   (2): Crm Cd 3, Crm Cd 4
## dttm  (1): DATE OCC
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#length(unique(df$`Crm Cd`)) # There is a total of 140 unique crime codes for part 1, this is useful fo

unique_codes <- unique(df[, c("Crm Cd", "Crm Cd Desc")])

unique_codes <- unique_codes[order(unique_codes$`Crm Cd`), ]

unique_codes
```

```
## # A tibble: 140 x 2
##    `Crm Cd` `Crm Cd Desc`
##       <dbl> <chr>
##  1      110 CRIMINAL HOMICIDE
##  2      113 MANSLAUGHTER, NEGLIGENT
##  3      121 RAPE, FORCIBLE
##  4      122 RAPE, ATTEMPTED
##  5      210 ROBBERY
##  6      220 ATTEMPTED ROBBERY
##  7      230 ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT
##  8      231 ASSAULT WITH DEADLY WEAPON ON POLICE OFFICER
##  9      235 CHILD ABUSE (PHYSICAL) - AGGRAVATED ASSAULT
## 10      236 INTIMATE PARTNER - AGGRAVATED ASSAULT
## # i 130 more rows
```

Since each crime incident can have more than one incident where crm code 1 is the primary, crm code 2-4 are additional crime codes, we must examin to make sure the the additional crimes aren't also violent.

```
# This checks if there are any na values for crm code 3, and all are na's. same thing with crm code 4. 
cat("There is a total of", sum(!is.na(df$`Crm Cd 1`)), "Non NA values for Crm Cd 1 \n")
```

```
## There is a total of 965933 Non NA values for Crm Cd 1
```

```
cat("There is a total of", sum(!is.na(df$`Crm Cd 2`)), "Non NA values for Crm Cd 2 \n")
```

```
## There is a total of 66210 Non NA values for Crm Cd 2
```

```
cat("There is a total of", sum(!is.na(df$`Crm Cd 3`)), "Non NA values for Crm Cd 3 \n")
```

```
## There is a total of 0 Non NA values for Crm Cd 3
```

```
cat("There is a total of", sum(!is.na(df$`Crm Cd 4`)), "Non NA values for Crm Cd 4 \n")
```

```
## There is a total of 0 Non NA values for Crm Cd 4
```

```
unique_codes_2 <- sort(unique(df$`Crm Cd 2`))
cat("Summary of unique codes in crime codes 2 section, \n")
```

```
## Summary of unique codes in crime codes 2 section,
```

```
unique_codes_2
```

```
##   [1] 210 220 230 231 235 236 250 251 310 320 330 331 341 343 345 350 351 352
##  [19] 354 410 420 421 430 431 433 434 437 438 440 441 442 445 450 451 470 471
##  [37] 473 474 480 485 486 510 520 521 522 623 624 625 626 627 647 648 649 651
##  [55] 653 654 660 661 662 664 666 668 670 740 745 753 755 760 761 762 763 810
##  [73] 812 813 814 815 820 821 822 830 846 850 860 865 886 888 890 900 901 902
##  [91] 903 906 910 920 921 922 924 928 930 931 932 933 940 942 944 945 946 948
## [109] 949 951 954 956 972 976 978 979 980 990 993 994 996 997 998 999
```

I just confirmed that only additional crime code 1 (primary) and 2 (secondary) slots contain crimes, while crime code 3 and 4 slots only have N/A values

Now I will filter the data into a new csv file containing only the violent crimes

```
library(stringr)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
violent_keywords <- c(
  "HOMICIDE", "MANSLAUGHTER", "RAPE", "SEXUAL",
  "ROBBERY", "ASSAULT", "BATTERY", "CHILD ABUSE", "DEADLY"
)



violent_codes_manual <- c(110, 113, 121, 122, 210, 220, 230, 231, 235, 236, 250, 251, 434, 435, 436, 437


violent_codes_keyword <- df %>%
  mutate(desc_upper = str_to_upper(`Crm Cd Desc`)) %>%
  filter(str_detect(desc_upper, paste(violent_keywords, collapse = "|"))) %>%
  pull(`Crm Cd`) %>%
  unique()


violent_codes <- union(violent_codes_manual, violent_codes_keyword)

violent_df <- df %>%
  filter(
    `Crm Cd`   %in% violent_codes |
    `Crm Cd 2` %in% violent_codes |
    `Crm Cd 3` %in% violent_codes |
    `Crm Cd 4` %in% violent_codes
  ) %>%
  mutate(year = year(`DATE OCC`))
```

Now we can begin analyzing 1. Get the sum of violent crimes per year 2. Trend analysis 3. Fit a linear regression model 4. Anova testing 5. Visualizations

1. Sum per year

```r
df_2020 <- violent_df %>% filter(year == 2020)
df_2021 <- violent_df %>% filter(year == 2021)
df_2022 <- violent_df %>% filter(year == 2022)
df_2023 <- violent_df %>% filter(year == 2023)
df_2024 <- violent_df %>% filter(year == 2024)
```

```r
sum_2020 <- nrow(df_2020)
sum_2021 <- nrow(df_2021)
sum_2022 <- nrow(df_2022)
sum_2023 <- nrow(df_2023)
sum_2024 <- nrow(df_2024)




cat("Total violent crimes in 2020:", sum_2020, "\n")
```

```
## Total violent crimes in 2020: 54259
```

```r
cat("Total violent crimes in 2021:", sum_2021, "\n")
```

```
## Total violent crimes in 2021: 68670
```

```r
cat("Total violent crimes in 2022:", sum_2022, "\n")
```

```
## Total violent crimes in 2022: 72306
```

```r
cat("Total violent crimes in 2023:", sum_2023, "\n")
```

```
## Total violent crimes in 2023: 71736
```

```r
cat("Total violent crimes in 2024:", sum_2024, "\n")
```

```
## Total violent crimes in 2024: 20777
```

There is a really low total number of violent crimes for 2024, hinting at an issue for 2024 data.

Quick check of total crimes commited per year

```r
df_2 <- df
cat("total crime in 2020:", nrow(df_2 %>% mutate(year = year(`DATE OCC`)) %>% filter(year == 2020)), "\n
```

```
## total crime in 2020: 160897
```

```r
cat("total crime in 2021:", nrow(df_2 %>% mutate(year = year(`DATE OCC`)) %>% filter(year == 2021)), "\n
```

```
## total crime in 2021: 209876
```

```r
cat("total crime in 2022:", nrow(df_2 %>% mutate(year = year(`DATE OCC`)) %>% filter(year == 2022)), "\n
```

```
## total crime in 2022: 235259
```

```r
cat("total crime in 2023:", nrow(df_2 %>% mutate(year = year(`DATE OCC`)) %>% filter(year == 2023)), "\n
```

```
## total crime in 2023: 232345
```

```r
cat("total crime in 2024:", nrow(df_2 %>% mutate(year = year(`DATE OCC`)) %>% filter(year == 2024)), "\n
```

```
## total crime in 2024: 127567
```

```r
cat("violent crimes in 2024:", sum_2024, "\n")
```

```
## violent crimes in 2024: 20777
```

There is an issue with data inconsistency for 2024. I did some research and found that the crime reporting system changed in 2024 so there is a lack of data. So we won't take 2024 into the analysis as it will skew the analysis.

2. Trend analysis

```r
# percentage change
percentage <- ((71736 - 54259) / 54259) * 100
cat("Rate change in violent crimes from 2020 to 2023:", percentage, "\n")
```

```
## Rate change in violent crimes from 2020 to 2023: 32.21032
```

```r
cat("Total violent crime in 2020:", nrow(df_2020), "\n")
```

```
## Total violent crime in 2020: 54259
```

```r
cat("Total violent crime in 2023:", nrow(df_2023), "\n")
```

```
## Total violent crime in 2023: 71736
```

```r
# ((new - old) / old) * 100

rate_2021 <- ((nrow(df_2021) - nrow(df_2020)) / nrow(df_2020)) * 100
rate_2022 <- ((nrow(df_2022) - nrow(df_2021)) / nrow(df_2021)) * 100
rate_2023 <- ((nrow(df_2023) - nrow(df_2022)) / nrow(df_2022)) * 100

average_rate <- sum((rate_2021 + rate_2022 + rate_2023)) / 3

cat("Average rate percentage increase per year from 2020-2023", average_rate, "\n")
```

```
## Average rate percentage increase per year from 2020-2023 10.35541
```

3. Fit a linear regression model where slope (positive = increasing violent crime)

p-value (is the trend statistically significant)

R^2 (how much of the variation is explained by year)

```r
df_yearly <- violent_df %>%
  mutate(year = year(`DATE OCC`)) %>%
  filter(year != 2024) %>%
  count(year, name = "violent_crime_count")

model <- lm(violent_crime_count ~ year, data = df_yearly)
summary(model)
```

```
##
## Call:
## lm(formula = violent_crime_count ~ year, data = df_yearly)
##
## Residuals:
##     1     2     3     4
## -4074  4731  2760 -3417
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11267201    4879534  -2.309    0.147
## year             5607       2414   2.323    0.146
##
## Residual standard error: 5397 on 2 degrees of freedom
## Multiple R-squared:  0.7296, Adjusted R-squared:  0.5943
## F-statistic: 5.395 on 1 and 2 DF,  p-value: 0.1459
```

Since the p-value is greater than 0.05, this means that the linear trend is not statistically significant, but this is primarly due to there being only 4 yearly observation, so this model doesn't have enough statistically power to declare the trend. This means 73% of the variation in yearly violent crime counts is explained by the year. The trend can't be declared statistically significant with the available data.

So instead I will perform one-way ANOVA 4. Anova testing on monthly basis

```r
df_monthly <- df %>%
  mutate(year = year(`DATE OCC`),
         month = month(`DATE OCC`)) %>%
  filter(year %in% 2020:2023) %>%
  count(year, month, name = "crime_count")

df_monthly$year <- factor(df_monthly$year)

anova_monthly <- aov(crime_count ~ year, data = df_monthly)
summary(anova_monthly)
```

```
##             Df   Sum Sq  Mean Sq F value   Pr(>F)
## year         3 90449169 30149723    30.9 1.04e-10 ***
## Residuals   42 40976661   975635
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(anova_monthly)
```

```
##   Tukey multiple comparisons of means
```

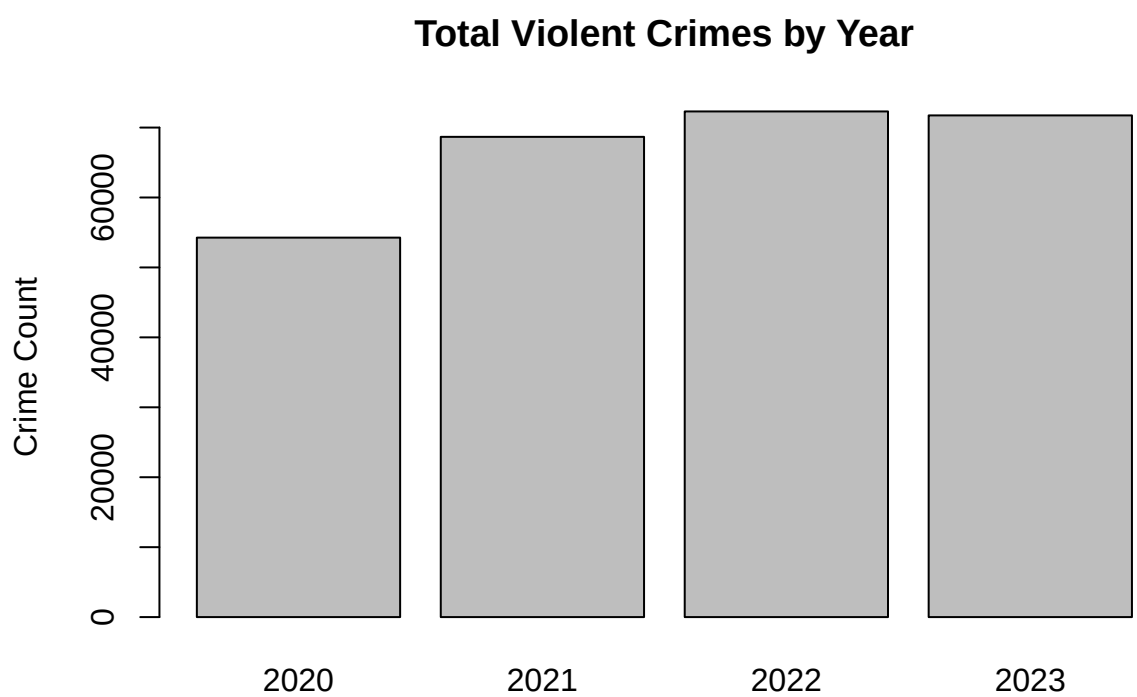
```
##     95% family-wise confidence level
##
## Fit: aov(formula = crime_count ~ year, data = df_monthly)
##
## $year
##                diff        lwr       upr     p adj
## 2021-2020 1399.9667   268.6578 2531.2755 0.0099851
## 2022-2020 3515.2167  2383.9078 4646.5255 0.0000000
## 2023-2020 3272.3833  2141.0745 4403.6922 0.0000000
## 2022-2021 2115.2500  1036.5893 3193.9107 0.0000277
## 2023-2021 1872.4167   793.7560 2951.0774 0.0001911
## 2023-2022 -242.8333 -1321.4940  835.8274 0.9307999
```

Since F-test is signficant due to $p < 0.05$, A one-way ANOVA showed a statistically significant effect of year on monthly violent-crime counts from 2020 to 2023. This indicates that average monthly violent-crime levels were not constant across years and that at least one year differed significantly from the others.

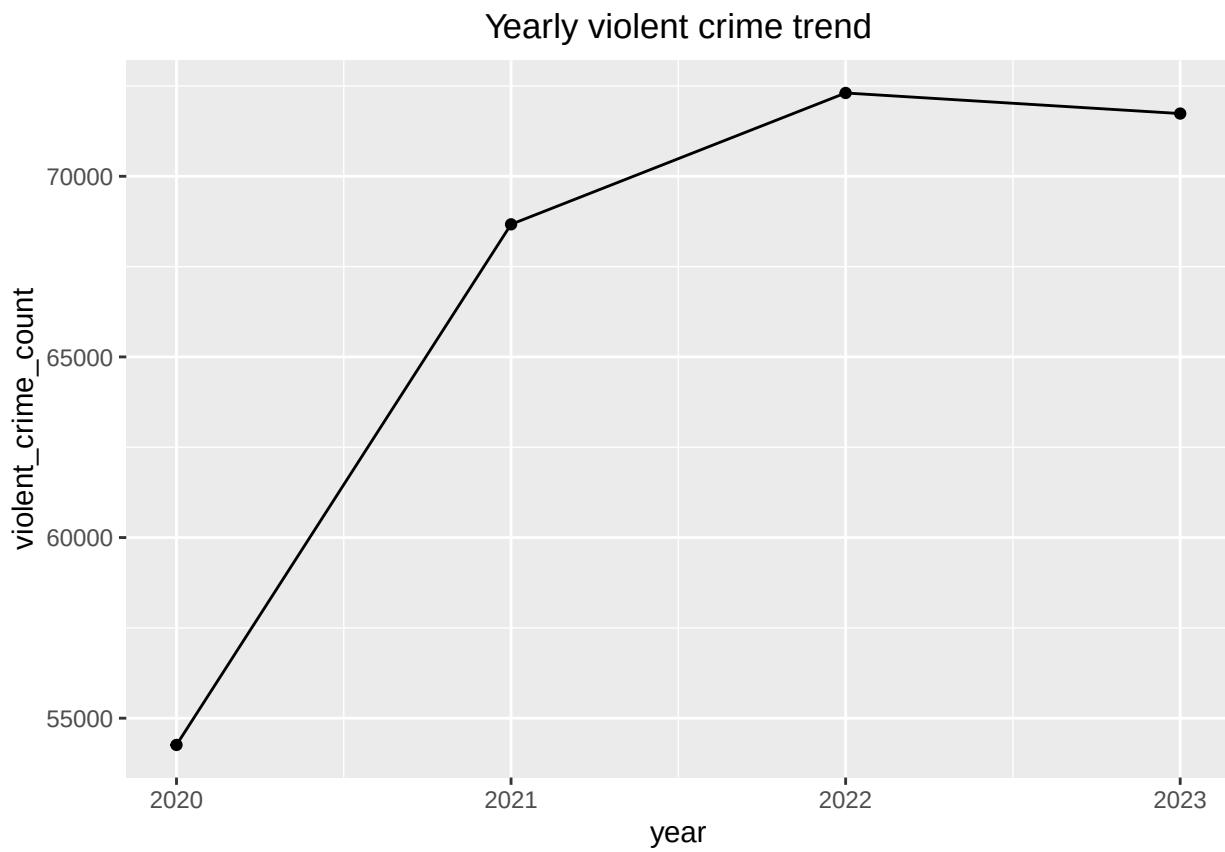
TukeyHSD results shows which years differ. Post-hoc Tukey comparisons revealed that 2021, 2022, and 2023 all had significantly higher monthly violent-crime counts than 2020. The increase from 2021 to 2022 was also significant, while the difference between 2022 and 2023 was small and not statistically significant.

5. Vizualization

```r
library(ggplot2)
summary <- c(sum_2020, sum_2021, sum_2022, sum_2023)
barplot(summary, names.arg = c(2020, 2021, 2022, 2023), main = "Total Violent Crimes by Year", ylab = "C
```

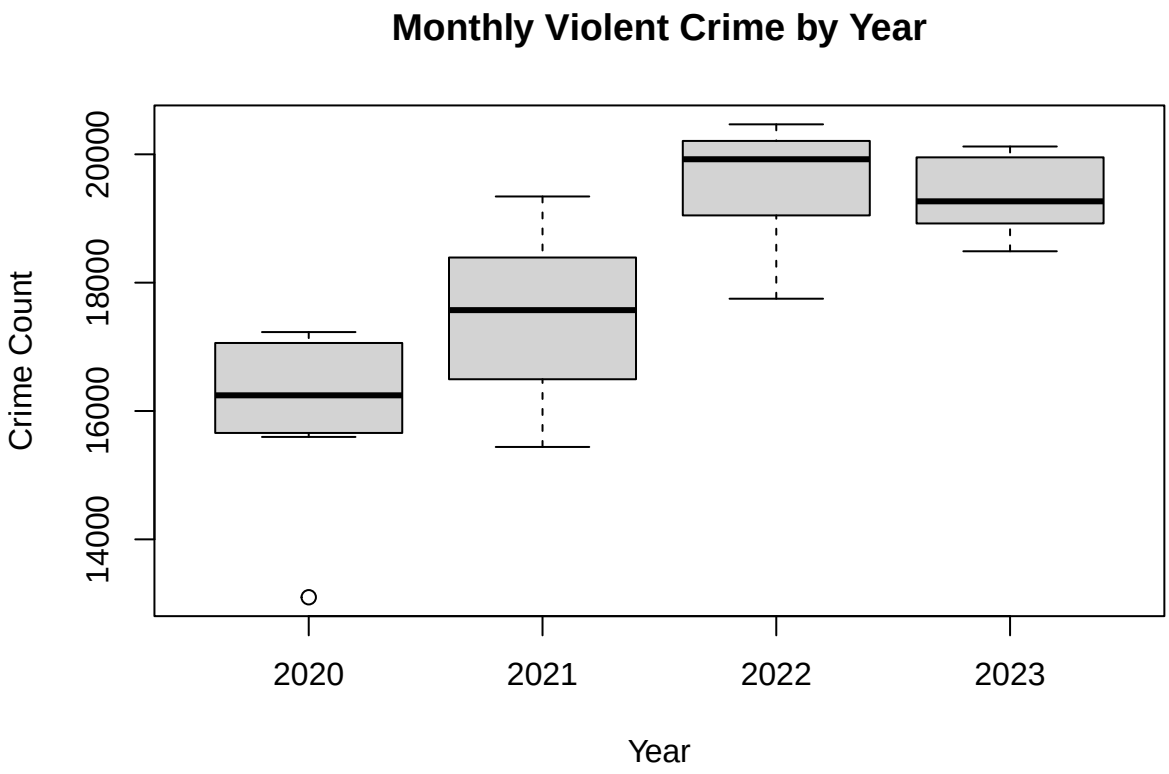


```r
# Year-over-Year Change Plot
ggplot(df_yearly, aes(year, violent_crime_count)) +
  geom_line() +
  geom_point() +
   labs(title = "Yearly violent crime trend") +
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Anova Box plots for monthly

boxplot(crime_count ~ year, data = df_monthly,
        main = "Monthly Violent Crime by Year",
        xlab = "Year", ylab = "Crime Count")
```



Graph 1) There is an increasing trend in crimes, as stated before. This is just a visual representation of this

Graph 2) Same conclusions as graph 1, just a different visualization

Graph 3) A box plot is a compressed summary of a distribution where each box shows the spread and center of monthly crime counts for a given year where the thick line inside the box is the median monthly crime count for that year. 2020 has a lower box, so lower crime levels. 2021-2023 have higher boxes so higher crime levels. 2022 has the highest box, so it has the highest monthly crime levels overall. The upward shift in the entire distribution across years visually supports the statistically significant differences found in the ANOVA and Tukey tests.

Results summary

1. The total count of violent crimes increase year by year.
2. There is an approximately 32% increase in crimes from 2020 to 2023, implying that there is an average rate percentage increase of approximately 10% a year.
3. The trend can't be declared statistically significant with yearly data.
4. A one-way ANOVA showed a statistically significant effect of year on monthly violent-crime counts from 2020 to 2023
5. The upward shift in the entire distribution across years visually supports the statistically significant differences identified.

Conclusion

There is a yearly upward trend in crimes, in the future it would be important to combine this data with real world situations that impact crimes such economy, education levels, media, presedntial status, etc..